

Pacifique Nsabumukiza

Computer Engineering Student · Embedded Systems · AI & Computer Vision · Junior Full-Stack Development

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PROFILE

Second-year Computer Engineering student at Kirikkale University. Founder of Paccy Developer delivering web, AI, and automation solutions to clients. Experienced in ESP32 firmware development, KiCad PCB design, YOLOv8 computer vision, and full-stack web development.

TECHNICAL SKILLS

Hardware & Embedded: ESP32 · Arduino · Raspberry Pi · C/C++ · I2C · SPI · UART · GPIO · PWM · KiCad PCB · Firmware Dev

AI & Computer Vision: YOLOv8 · OpenCV · Pytorch · scikit-learn · NumPy · Pandas · Google Colab · Ultralytics

Web & Software: Python · C# · JavaScript · React · HTML5 · CSS3 · BOOTSTRAP · ASP.NET Core · MySQL · Next.js · TypeScript · Tailwind CSS · Octave

Tools: n8n · Git · GitHub · Figma · Photoshop · Canva · Arduino IDE · VS Code · KiCad · Blender · Fusion360 · WordPress

EXPERIENCE

Founder & Lead Developer · Paccy Developer 2025 – Present

Kirikkale, Turkey

- Delivered web and AI projects for **5+ clients**, managing full lifecycle from scoping to deployment.
- Built **Next.js 14 / TypeScript / Tailwind CSS** crypto dashboard (Crypgo) with live REST API integration and 40% faster load times via WebP asset optimization.
- Automated a 2-hour daily research task by building an **n8n + LLaMA + SerpAPI** pipeline delivering results to a Telegram bot.

Embedded Software Developer 2024 – Present

Kirikkale, Turkey

- Designed a **4-layer KiCad PCB** for a 20A industrial smart home controller built around **ESP32-WROOM-32**, handling Wi-Fi/Bluetooth control of high-power loads via a RAYEX-L90AS relay.
- Wrote **ESP32 firmware in C++** configuring GPIO, PWM, and UART for sensor acquisition and motor control in an autonomous harvesting robot.
- Designed and DRC-validated **KiCad PCB schematics** for an AC signal measurement circuit (half-wave rectifier + RMS sampling loop).
- Implemented **I2C and SPI** sensor interfaces with interrupt-driven acquisition and sleep-mode scheduling to reduce power draw.

Web Developer 2023 – Present

Kirikkale, Turkey

- Delivered **5+ client websites** in React, HTML5, and CSS3 with mobile-first layouts and cross-browser compatibility.
- Building **EduCore** school management system using ASP.NET Core, Blazor Server, MySQL, and Entity Framework Core.
- Built and deployed personal **developer portfolio** (paccywebdev.github.io) using HTML, CSS, and Bootstrap . Optimised assets from PNG to WebP/JPG, cutting load time by 30%.

Graphic & UI/UX Designer 2022 – Present

Bujumbura, Burundi → Kirikkale, Turkey

- Designed brand identities and UI prototypes in **Figma and Photoshop**; translated mockups directly to Tailwind CSS production code.

PROJECTS

AI Automation Assistant (Paccy AI) 2026

- Built local AI assistant using **n8n** + **OpenRouter (LLaMA)** + **Simple Memory** + **SerpAPI**, serving answers through a Telegram bot with zero-downtime hosting.

Autonomous Plastic Detection System · Kırıkkale University **2025**

- Trained **YOLOv26n** on a Plastic detection dataset, reaching **99.5% mAP@50** via Ultralytics on Google Colab.
- Designed v2 architecture with **LiDAR depth estimation**, **OpenCV optical flow**, instance segmentation for damaged fruit, and **MQTT telemetry** logging.

Numerical Analysis Library · Kırıkkale University **2025**

- Co-built root-finding library in **C#** and **Python** implementing Euler, Bisection, Newton-Raphson, Secant, Gaussian Elimination, and Cramer's Rule with error analysis.

Competitive Queuing Neural Network (Octave) · Tujenge Scholars Program **Mar 2023 – May 2023**

- Implemented a **Competitive Queuing Neural Network in Octave** to simulate **chimpanzee brain activity**, and self-inhibitory mechanisms across sequence, competitive, and output layers.

EDUCATION

BSc Computer Engineering · Kırıkkale University **2023 – 2028**

Kırıkkale, Turkey

University Preparation · Tujenge Scholars Program **2022 – 2023**

High School Diploma · Lake Tanganyika High School **2015 – 2021**

Burundi

CERTIFICATIONS

- IBM AI Essentials
- Machine Learning (BTK AKADEMI)
- Turkish Language C1
- SQL (Turkcell geleceği yazarlar)
- AI & Machine Learning (Data Science & AI Essentials)
- Python & Data Analysis
- Kaggle Badge (Data Science with Python)
- Google Developer Student
- Bioinformatics & Data Processing
- Cloud & Computing

LANGUAGES

English (Fluent) · French (Fluent) · Turkish (C1) · Kirundi (Native) · Kinyarwanda (Fluent) · Swahili (Fluent)